FALL 2024

CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 8 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

By,

Eric Nguyen and Daniel Mccomas

Under the guidance of

Dr. Jason Losh

Introduction

In this lab we will be attempting to implement a LIDAR (Light Detection and Ranging) sensor to obtain the area of a room. A LIDAR sensor is a remote sensing technology that uses laser light to measure distances and create high-resolution maps of environments. LIDARs are widely used in fields like autonomous vehicles, robotics, environmental monitoring, and mapping.

Our sensor works by sending a beam of light and waiting for the response time to obtain distance. A motor is attached to the sensor for it to spin and get a 360-degree scan of the room. From our formula and the polar plot, we can obtain the area of the room.

Methods

1. First, we connected the A1M8 LIDAR assembly using 5 cables to the white board. We applied power to the 5V input (RED) from Vbus and ground (BLACK) from GND. We connected the PWM input (WHITE) to a GPO to enable the motor.
2. We powered on the TM4C123GXL board and output a short pulse on PWM to turn the motor on and then off again.
3. We connected the TX output (BLUE) to U1RX and the RX input (VIOLET) to U1TX.
4. We wrote code to initialize UART1 to 115,200 baud, 8N1.
5. We sent a STOP command to the LIDAR sensor.
6. We sent an INFO command to the LIDAR sensor, verified that the information in the response was as expected, and displayed it on UART0.
7. We turned on the motor and waited until it reached the correct speed.
8. We sent a SCAN command (or an advanced version) to the LIDAR sensor.
9. We verified that the response was as expected.
10. We entered an endless loop to read the bearing and distance data from the sensor until a full data set was created (one full rotation of data).
11. We sent a STOP command to the LIDAR sensor and turned off the PWM signal to stop the motor.
12. We calculated the total room area on the TM4C123GXL board and displayed it on UART0.
13. Lastly, we output the data set in a tabular format on UART0, took the terminal data, imported it into a spreadsheet, and displayed it as a polar plot. We verified that the image was as expected, noting that the window and open door appeared a bit unusual at first in the scans.

The data sheet had all of the required information that was needed to implement the lidar sensors. The only thing we did differently was that we applied voltage straight to the motor for it to constantly spin instead of using PWM. A description of some of the process that we had to do to get the information will now be discussed. All commands start with the same hex digits of A5 followed by the command and then descriptors. For example, we would transmit on the TIVA board's UART TX line

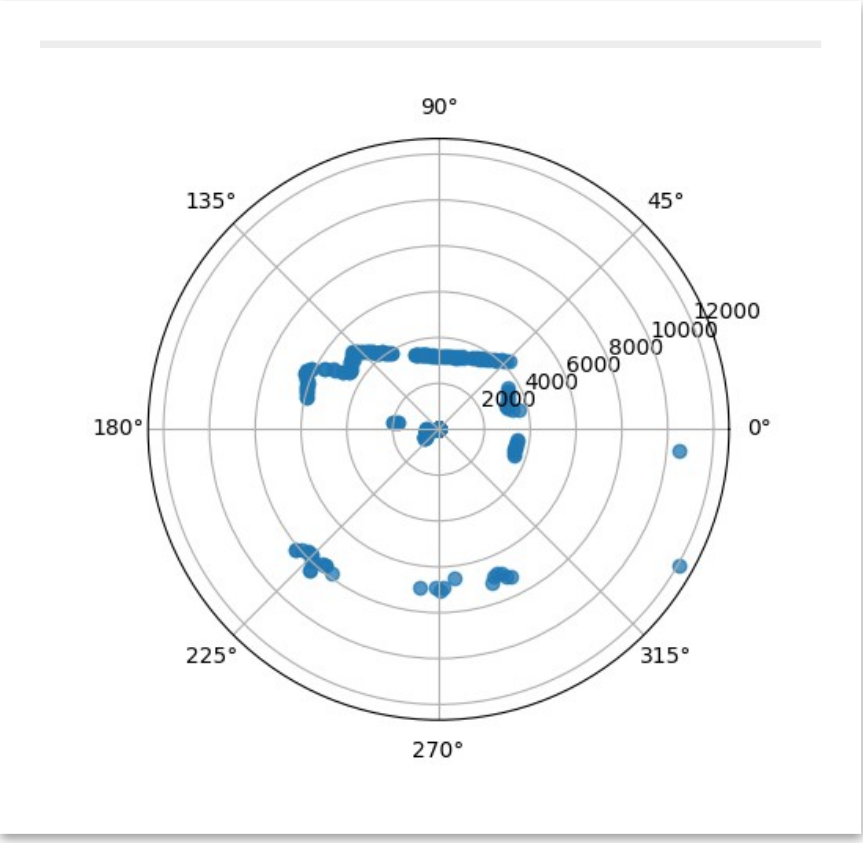
0x65 followed by 0x20 (scan command) to start a scan. The UART RX line on the tiva would then return the descriptor bits and then data generated by the Lidar.

In the case of a scan command, we are receiving data in a raw format. We would then convert these raw values into angle and degree values so we can map the room and compute the square inch area. With some data processing and triangulation formulas we were able to get an area reading. The triangulation formula was summation((1/2)(distance1*distance2)sin(theta2-theta1)). This reading was not always consistent but often would give us a relatively close value. The variation could have a lot of sources, but in the end it doesn't neccessarily matter for the general purpose of learning how a lidar works in this lab.

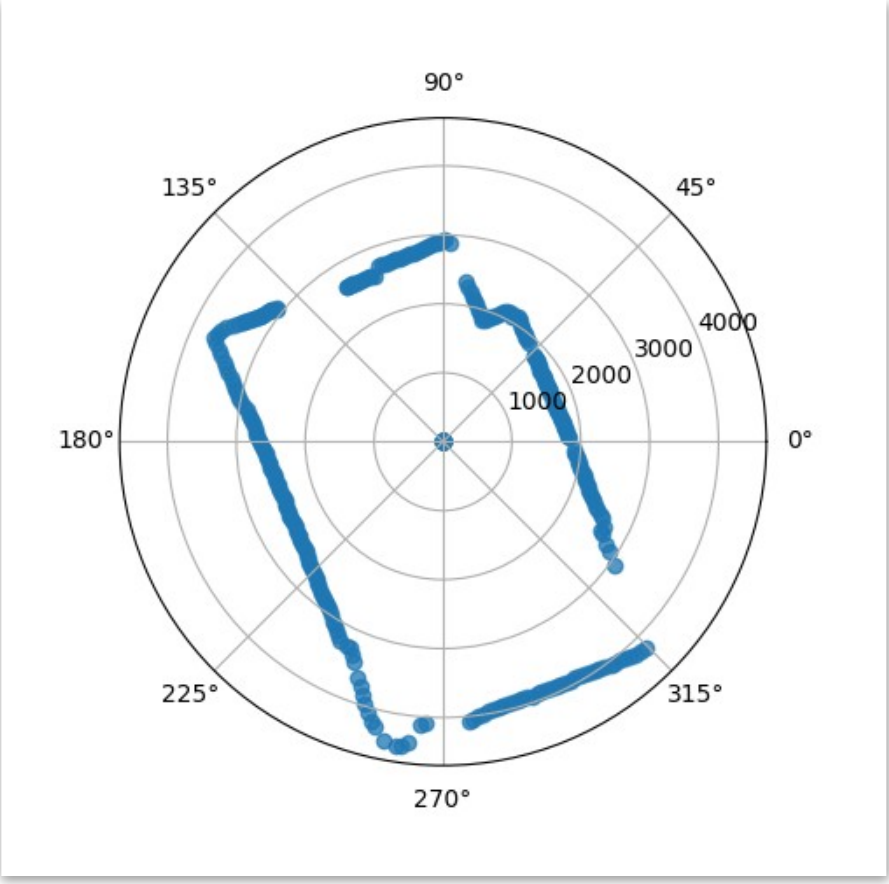
Initial Experiments

After running the program, we took the data and used a polar plot to verify if the lidar sensor was working.

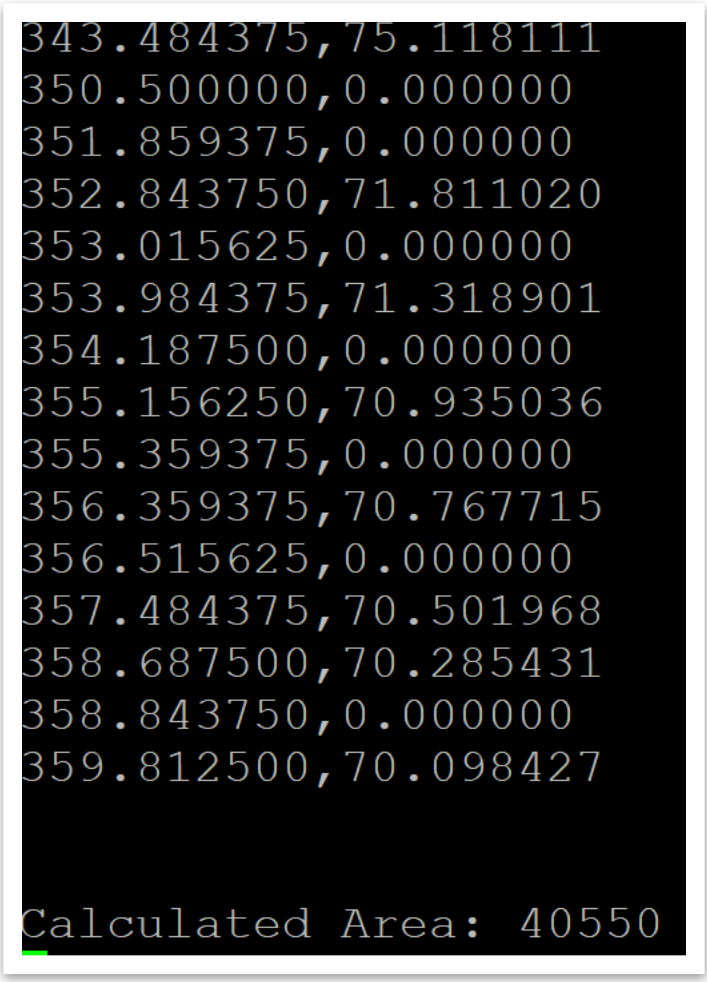
Polar Plot of Room ERB 335:



Polar Plot of Room ERB 122:



This is a small output on the terminal to receive area:



Conclusion

In conclusion, we were able to implement the LIDAR sensor to find the area of a room. LIDARs have many advantages because they implement high precision and accuracy and the ability to operate in various lighting conditions. In a dark room, with minimal movement we were able to get consistent readings but in a room with many people the values jumped around quite a bit but still was within reason. Some issues we had were that the hardware we used wobbled and made the polar plots inconsistent. Small boxes used for testing also were an issue because of the dead zones of the LIDAR. Overall however, this lab was a great introduction into the world of LIDAR.

Circuit Built

